

TSE-CNN: A Two-Stage End-to-End CNN for Human Activity Recognition

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Abstract—Human activity recognition has been widely used in healthcare applications such as elderly monitoring, exercise supervision, and rehabilitation monitoring. Compared with other approaches, sensor-based wearable human activity recognition is less affected by environmental noise and therefore is promising in providing higher recognition accuracy. However, one of the major issues of existing wearable human activity recognition methods is that although the average recognition accuracy is acceptable, the recognition accuracy for some activities (e.g., ascending stairs and descending stairs) is low, mainly due to relatively less training data and complex behavior pattern for these activities. Another issue is that the recognition accuracy is low when the training data from the test subject are limited, which is a common case in real practice. In addition, the use of neural network leads to large computational complexity and thus high power consumption. To address these issues, we proposed a new human activity recognition method with two-stage end-to-end convolutional neural network and a data augmentation method. Compared with the state-of-the-art methods (including neural network based methods and other methods), the proposed methods achieve significantly improved recognition accuracy and reduced computational complexity.

Index Terms—Human activity recognition, motion sensor, neural network, end-to-end.

I. INTRODUCTION

HUMAN activity recognition (HAR) is used to detect and classify human activity through automatic analysis by machine. It has been widely adopted in healthcare applications, such as elderly monitoring, exercise supervision and rehabilitation monitoring [1]–[7], to maintain or improve health. For example, it can be used to detect and analyze the activities of the elderly to predict potential health issues. It can also be used to monitor people's daily exercise and provide suggestions for them to exercise in a healthier way. Besides, it can be used for

activity analysis for post-operative rehabilitation patients to obtain a better understanding of their current state, so as to assist physicians in the assessment and treatment of the patients.

HAR can be divided into two categories—vision-based HAR and sensor-based wearable HAR. Vision-based HAR is low cost and easy to implement, but it is affected by environmental noise such as lighting, shooting angles and overlap between subjects, making it difficult to improve the classification accuracy [8]. Compared to the vision-based HAR, sensor-based wearable HAR is not affected by the aforementioned noise and is promising to achieve higher accuracy, which has attracted lots of attention from research community. In this work, we mainly focus on sensor-based wearable HAR.

A sensor-based wearable HAR works as follows. It first collects data from multiple motion sensors, such as accelerometers, gyroscopes and magnetometers, which are placed at waists, thighs, ankles, wrists, etc. [9]–[12]. Then, it performs data pre-processing such as de-noising and data normalization. After that, feature extraction and classification are performed to extract the features and classify the activities (e.g., walking, running, ascending stairs, descending stairs, crank arms, etc.). In the past, different feature extraction and classification methods have been proposed. For the feature extraction, various types of manual features have been investigated such as mean value, variance, covariance, skewness, frequency component and wavelet coefficient [4], [8], [13], [14]. For the classification, conventional machine learning methods [13]–[16] and neural network methods [17]–[19] have been investigated. However, there are some issues in the manual feature extraction. First, manually extracted features are limited by human knowledge and experience, resulting in limited accuracy. Second, to achieve satisfactory recognition accuracy, a large number of features are required which increases the computational complexity significantly. Third, it takes a lot of time and effort to find good features to achieve high recognition accuracy. To address the above issues, recently many researchers propose end-to-end neural network to classify human activities [20]–[24]. The end-to-end neural network (i.e., no hand-designed feature extraction is involved) can learn features automatically, avoiding the deficiencies of human knowledge and eliminating the hassle of finding suitable features. However, one of the issues of the existing methods is that although the average accuracy is acceptable, the accuracy for certain activities such as walk downstairs and upstairs is still very low [22], [25]–[27]. In addition, the recognition accuracy is heavily affected by the amount of training data. When the amount of training

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data is small, the accuracy is low. Moreover, the computational complexity for neural network is still high which increases the power consumption, making it unsuitable to wearable devices.

In this work, to address those issues, we propose a new end-to-end HAR method. Using data from a single accelerometer, the proposed method significantly improves the recognition accuracy for ascending stairs and descending stairs activities which has low accuracy in the conventional HAR methods, while maintaining a high accuracy for other activities. It also helps reduce the computational complexity and power consumption. Moreover, we propose a data augmentation method, which helps improve the recognition accuracy when the amount of test subject's data is limited.

The rest of paper is organized as follows. Section II reviews related works on HAR. Section III describes the details of the proposed methods. Section IV gives the experimental results. Section V gives the benchmark table and discussions and Section VI concludes the paper.

II. RELATED WORK

This Section reviews the existing work on sensor-based wearable HAR. We firstly introduce methods based on feature extraction. [18] used data from an accelerometer placed at the waist for activity recognition. A total of 52 subjects' (39 males, 13 females, aged 21–64 years, height ranged between 1.53 and 1.88 m and weight between 42 and 94 kg) data is collected and the sampling rate is 50 Hz. Several features such as the sum of squares, standard deviations of wavelet coefficients are extracted and fed into an Multi-Layer Perceptron (MLP) for classifying five activities (i.e., walking flat, walking slope-up, walking slope-down, walking stairs-up and walking stairs-down). [28] performed activity recognition using data from an accelerometer and a gyroscope at a sampling rate of 50 Hz. A group of 30 volunteers with ages ranging from 19 to 48 years were selected for this experiment. There were six activities such as walking, sitting, standing. A 561-dimensional feature vectors were extracted and one-against-all support vector machine (OAA SVM) was used to improve the recognition accuracy. [14] considered the transition activity and the unknown activity. They extracted a 561-dimensional feature vector based on [18], [28] and used the improved SVM to achieve higher recognition accuracy. [8] used data from an accelerometer and a gyroscope in a smartphone like [14]. A total of 10 male subjects, whose heights range from 165 cm to 178 cm, weights range from 55 kg to 72 kg and aging from 20 to 23 were selected in their experiment. The sampling rate is 20 Hz and total five activities (i.e., descending stairs, ascending stairs, walking, jogging and jumping) are recognized in the paper. They considered the placement of the smartphone in trousers pocket, jacket pocket, hand, upper arm and waist to recognize five activities. In total 89-dimensional feature vectors are extracted and several classification methods including SVM, k-Nearest Neighbor (KNN) and random forest (RF) are compared under different modes. [19] and [15] used data from an accelerometer in a smartphone placed in the pants pocket to classify six behaviors (i.e., walking, jogging, standing, sitting, descending stairs and ascending stairs). [19] extracted a 43-dimensional feature vector and used MLP for activity classification.

[15] used an automatic encoder to reduce the dimensions of the feature vectors from [19] and achieved higher recognition accuracy. [29] introduced an online sequential extreme learning machine to tackle the cross-location HAR problem based on statistical and frequency-domain features. It is able to recognize eight activities (i.e., walking (3 km/h), walking (5 km/h), running (6 km/h), running (12 km/h), ascending stairs, descending stairs, bicycling and rope jumping) using an accelerometer placed at ankle/waist/wrist with improved recognition accuracy. [30] focuses on addressing the problem of different orientations and different placements of an accelerometer embedded in the smartphone and different subjects. They proposed the coordinate transformation & principal component analysis and online SVM to recognized five activities (i.e., static, walking, running, going upstairs and going downstairs) using an accelerometer of a smart phone placed in pants' pocket, shirt's pocket and backpack. [6] employed a novel windowing method to efficiently segment data from a 3-axis accelerometer to reduce the power consumption. They extracted three features (i.e., mean, standard deviation and number of peaks of the measurements) for HAR. [17] used data from three accelerometers placed at wrist, hip and ankle respectively for HAR. To reduce the data dimension, intensity is calculated from raw accelerometer data and fed into multiple parallel neural networks for recognizing six activities (i.e., resting, typing, gesticulating, walking, running and cycling).

All the above work adopts feature engineering in the HAR. However, the accuracy of manual feature extraction is limited by human knowledge and experience. Another issue is, most of them require a large number of manually extracted features, which significantly increase the computational complexity. Also, the feature engineering takes a lot of human effort for finding proper features to achieve high recognition accuracy. For these reasons, many researchers have turned their attention to the end-to-end neural network, which is able to learn features automatically. The end-to-end methods are reviewed as follows.

[22] proposed an end-to-end 1-D CNN and used three datasets to compare the effectiveness of proposed method with KNN using manually extracted features and showed that the end-to-end CNN has higher accuracy. [31] used two dataset to compare an 1-D end-to-end CNN with other methods such as SVM and 1NN. The experimental results show that the 1-D end-to-end CNN achieves a high recognition accuracy. They also proposed a smoothing technique combined with 1-D CNN which further improves the accuracy. [24] compared different structures of end-to-end CNN for recognizing six activities (i.e., walking, walking upstairs, walking downstairs, standing, sitting and lying) using the data from a accelerometer and a gyroscope. [32] proposed a two-level CNN structure with one CNN in the first level and two CNNs in the second level. The CNN in the first level is used to classify activities into static activities and dynamic activities and the two CNNs in the second level are used to further classify the static and dynamic activities. [33] transformed and grouped the raw sensors data into a 2D activity image. Then the activity image is fed into a classifier consisting of 2-D CNN and SVM for activity recognition. [34] proposed a structure combining 2-D CNN and LSTM which improves the recognition accuracy for activities such as open

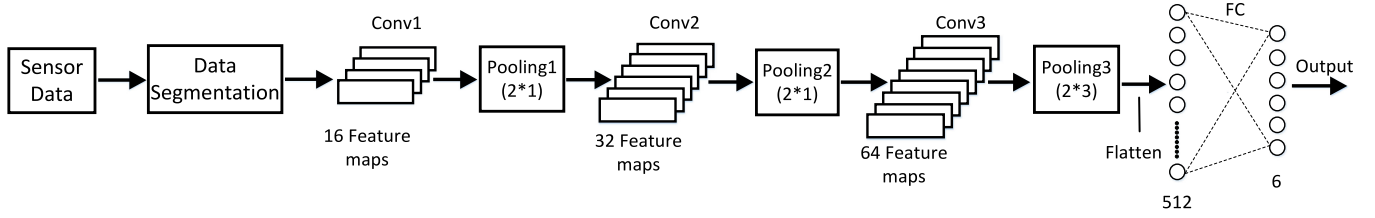


Fig. 1. The system architecture of the baseline method.

door and close door. [20] used an end-to-end 2-D CNN for activity recognition. Different convolution kernels are used for accelerometer and gyroscope data respectively, which improves the accuracy of recognition. However, this work uses 5 sensors (3 accelerometers and 2 gyroscopes) placed at different positions of human body (chest, right wrist and left ankle) which increases the cost and inconvenience for wearing. [21] used data from an accelerometer in a smart wristband placed in the wrist to recognize upper limb movements such as reach and retrieve an object, lift cup to mouth and perform pouring or unlocking action. They categorize the upper limb movements into three basic classes and used an end-to-end CNN structure for recognition. [23] used four sensors (an accelerometer, a gyroscope, a magnetometer and a barometer) in a smart phone carried in hands to classify eight activities (i.e., still, walking, running, false motion, walking upstairs, walking downstairs, up elevator and down elevator). They used an autoencoder to extract features and used a softmax layer for recognition, which improves accuracy.

The end-to-end neural network avoids the limitations of using feature engineering and improve the recognition accuracy. However, a common issue of existing end-to-end method is that although the average recognition accuracy is acceptable, some activities with relatively less training data and complex behavior patterns has low recognition accuracy (e.g., ascending stairs, descending stairs) [15], [19], [22]. Another issue is, for the personalized mode, some data from the test subject (i.e., user) is available for training to improve the recognition accuracy, however, the improvement is limited when the training data from the test subject is limited which is a common case in real practice. Finally, the end-to-end neural network still have high computational complexity, which leads to high power consumption, making it unsuitable for wearable devices.

III. PROPOSED TWO-STAGE END-TO-END CNN BASED HAR METHODS

In this section, we will introduce our proposed methods, including a two-stage end-to-end convolutional neural network (TSE-CNN) and a data augmentation method. In this work, six human activities are classified in total, including walking, jogging, standing, sitting, ascending stairs, descending stairs. Unlike other works that use data from multiple sensors including accelerometer, gyroscope and magnetometer, we propose a low cost solution with a single accelerator for the activity recognition. As the number of sensors increases, the data for processing increases significantly. This will not only increase the device cost, but also increase the processing time and power

TABLE I
THE PARAMETERS OF THE BASELINE

Layer	Size	Channel	Parameters
Input	64*3	1	-
Convolution1	15*3	16	736
Pooling1	2*1	16	-
Convolution2	2*2	32	2080
Pooling2	2*1	32	-
Convolution3	2*2	64	8256
Pooling3	2*3	64	-
Fully-Connected	512*6	-	3078

consumption. There are also some works show that HAR with a single accelerometer is feasible [2], [6], [22], [29]. For comparison, we have also built a baseline single-stage end-to-end CNN for the activity recognition. Next, we will introduce the baseline method and our proposed method.

A. Baseline

As described in Section I, conventional HAR using feature engineering suffer from several limitations, including accuracy limited by human knowledge and experience, large amount of features increasing the computational complexity, significant time and effort spent on finding proper features. To address these issues, end-to-end methods have been proposed. It is able to learn features automatically, no longer limited by human knowledge and experience, nor require significant time and effort for finding proper features. This does not only improve the recognition accuracy, but also reduce the computational complexity caused by the extraction of a large number of features. In this work, we designed our baseline by following an end-to-end single-stage CNN proposed recently in [20], [21]. This structure shows relatively good recognition performance among the state-of-the-art work. We have also optimized the baseline structure carefully by extensive experiments. The baseline has 3 convolution layers, 3 pooling layers and 1 fully connected layer, as shown in Fig. 1. Each convolution layer is followed by a pooling layer. After 3 convolution layer and 3 pooling layer, data is flatten and sent to a fully connected layer for final classification. The details of the baseline structure is shown in Table I.

B. Proposed TSE-CNN Method

As aforementioned, the conventional end-to-end methods (i.e., the baseline) avoid the limitations of manual feature extraction. However, the major issues of this method is although the average recognition accuracy is acceptable, the recognition

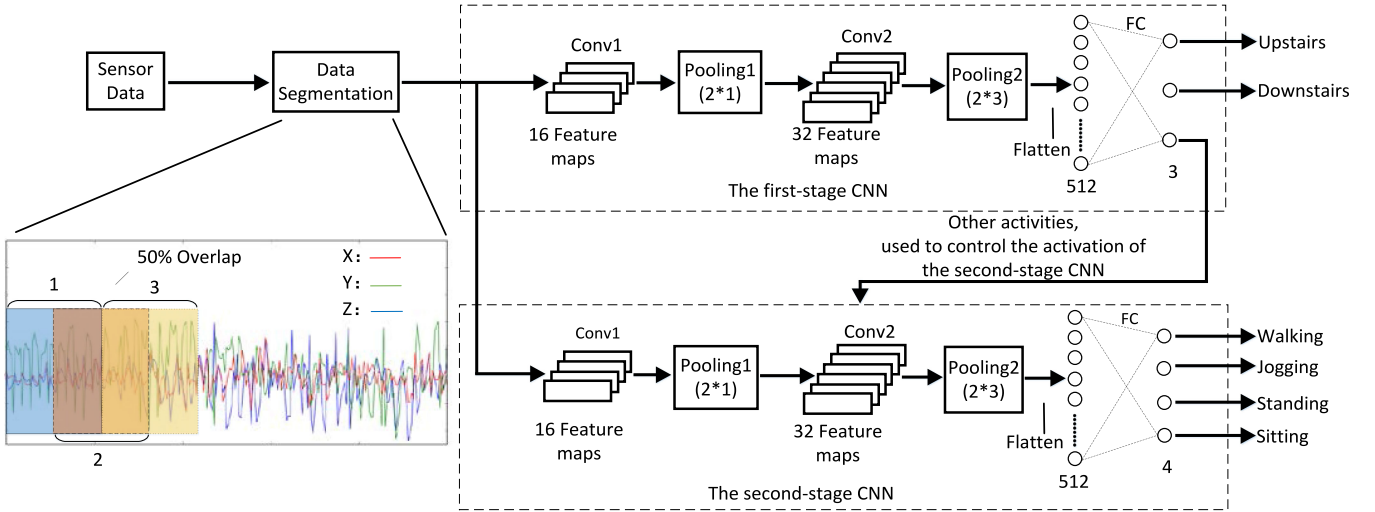


Fig. 2. The system architecture of the proposed end-to-end HAR method.

TABLE II
THE PARAMETERS OF TSE-CNN

Layer	Size	Channel	Parameters
Input	64*3	1	-
First-Stage:	-	-	-
Convolution1	15*3	16	736
Pooling1	2*1	16	-
Convolution2	2*2	32	2080
Pooling2	2*3	32	-
Fully-Connected	512*3	-	1539
Second-Stage:	-	-	-
Convolution1	15*3	16	736
Pooling1	2*1	16	-
Convolution2	2*2	32	2080
Pooling2	2*3	32	-
Fully-Connected	512*4	-	2052

accuracy for some activities (e.g., ascending stairs and descending stairs) is very low, mainly due to relatively less training data and complex behavior pattern for these activities. Another issue is that usually a relatively large CNN is required to achieve high accuracy, resulting in large computational complexity and thus high power consumption. To address these issues, we proposed a two-stage end-to-end CNN structure: TSE-CNN.

Fig. 2 shows the proposed TSE-CNN. After collecting data from an accelerometer, we first segment data using a sliding window with a window size of (64, 3), where 64 refers to the time points corresponding to a window of 3.2 s at a sampling frequency of 20 Hz, which is sufficient to cover most of the activity information without introducing too much delay, and 3 refers to tri-axis of the accelerometer. The window has an overlap of 50%. Then we use the TSE-CNN to perform the end-to-end activity classification. The proposed TSE-CNN consists of two cascaded CNNs. Both of them has only 2 convolution layers, 2 pooling layers and 1 fully connected layer, which is smaller than the CNN in the baseline structure (the detailed parameters of each layer are shown in Table II). The first-stage CNN focuses on classifying ascending stairs and descending stairs activities while the second-stage CNN focuses on classifying

the other activities. The output of the first-stage CNN has three classes: ascending stairs, descending stairs and other activities, while and the output of the second-stage CNN has four classes: walking, jogging, standing, sitting.

In the conventional end-to-end method, all the activities are classified with a large single-stage neural network. Among the 6 activities, it has been found that ascending stairs and descending stairs has the lowest accuracy mainly due to relatively less training data and complex behavior pattern. To address this issue, in the proposed TSE-CNN, we use a separate CNN stage to differentiate ascending/descending stairs and other activities. As this CNN stage is dedicated for the recognition of these two activities, it is able to achieve much higher accuracy than the baseline which uses a single-stage CNN to classify 6 activities. Another benefit of this structure is that, for the recognition of ascending stairs and descending stairs activities, only the first-stage CNN need to be activated and the second-stage CNN is inactivated, which significantly reduces the computational complexity and thus power consumption.

In the training phase, we first train the first-stage CNN to classify three activities: ascending stairs, descending stairs and other activities. Then we remove the training data of ascending stairs and descending stairs and train the second-stage CNN to classify the other four activities: walking, jogging, standing, sitting. In the inference phase, the data is fed into the first-stage CNN to judge whether this is ascending stairs, descending stairs or other activities. If it is ascending/descending stairs, the first-stage result is directly outputted as the final result. Otherwise, the data is fed into the second-stage CNN for further classification among walking, jogging, standing and sitting.

C. Proposed Data Augmentation Method

In HAR, the amount of training data from the test subjects (i.e., users) plays a critical role in the accuracy of HAR [21], [26], [35]. In real practice, only a small amount of training data from the test subjects is available, which heavily affect the accuracy of HAR. To address this issue, we propose a data augmentation method. The details of the method is shown in Fig. 3, which

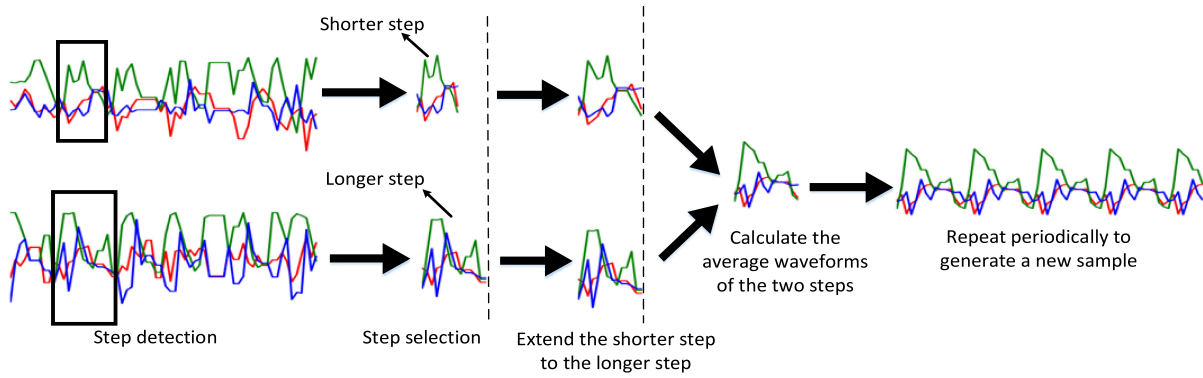


Fig. 3. Proposed data augmentation method.

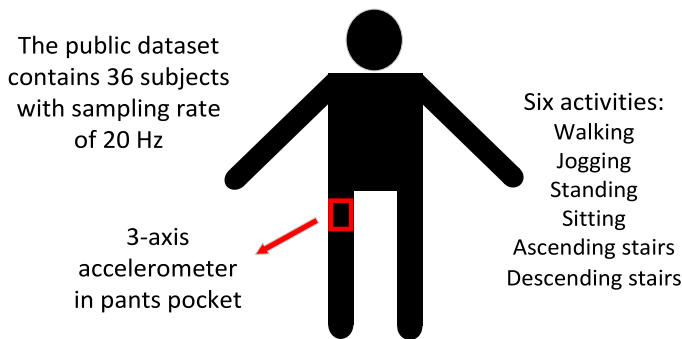


Fig. 4. The placement of the smartphone.

can be divided into 5 stages: 1) We select two samples (i.e., two windows of 64 points) belonging to the same activity of the test subject and perform step detection in the two samples to obtain two steps (i.e., periodic cycles). The way to obtain the two steps is to select the first step (i.e., periodic cycle) from each sample. 2) As the two steps obtained may have different length (i.e., containing different number of points), we extend the shorter one to the same length of the longer one for later processing by linear interpolation. For example, the shorter step contains 20 points and the longer step contains 35 points. Then 15 points need to be inserted into the shorter step by linear interpolation. 3) Then we generate the waveform of a new step by averaging the waveform of the two steps. 4) Then we repeats the newly built step waveform periodically to generate a new sample of 64 points. 5) Finally we apply stages 1–4 repeatedly to obtain more sample data.

Data augmentation can generate more artificial training data for the test subject so as to address the issue that the training data from the test subject is limited. This will help improve the recognition accuracy, as can be seen in the experimental results in Section IV.

IV. EXPERIMENTAL RESULTS

In this work, we use the public dataset WISDM [19] for the evaluation of the proposed methods. In this dataset, the data is sampled from a 3-axis accelerometer in a smartphone placed in pants pocket (shown in Fig. 4) with a sampling rate of 20 Hz. This dataset contains a total of 36 persons and 6 classes of

activities: walking, jogging, standing, sitting, ascending stairs and descending stairs. As the window contains 64 points (for each of the three axis), at a sampling rate of 20 Hz the time scale of the window is 3.2 s.

Two test modes are adopted to evaluate the proposed methods, including:

Rest to One (RTO): In this mode, the data of the test subject is excluded from the training data.

With Test Subject's Data (WTOD): In this mode, a small amount of the tested subjects' data (1/8) is included in the training data.

For testing, all of the 36 persons' data from the database are used. For each experiment, 5 persons' data are chosen as the test data and the remaining 31 persons' data is used as the training data. Following this method, 7 cross-validation experiments are performed. In each experiment, 5 persons with 6 classes of activities for each person are tested (i.e., test subjects). Between RTO and WTOD modes, the only difference is that in WTOD mode 1/8 of the test subject's data is included in the training data. Like most of other work, F-score is used to evaluate the effect of the proposed methods in the experiments.

A. Results for RTO Mode

Table III shows the F-score of each activity of the baseline method for 7 cross-validations in RTO mode. The test ID refers to the ID of the 5 test subjects in the cross-validation. Overall, the F-score of ascending stairs and descending stairs are much lower than the F-score of others, indicating that ascending stairs and descending stairs are more difficult to classify. In addition, for different people, the results are different. For example, the walking F-score of the test subjects 31–35 is about 13% higher than that of test subjects 26–30, meaning that the walking of the test subjects 31–35 is easier to recognize. Table IV shows the improvement of the proposed TSE-CNN method over the baseline. Across different test subjects, the proposed method has a large improvement over the baseline especially for ascending stairs and descending stairs (up to 6.2% and 11.3% for ascending stairs and descending stairs respectively, as shown in Table IV). In addition to the improvement in the F-score, the proposed method also reduces the amount of computation when classifying ascending stairs or descending stairs, as the second-stage CNN need not be activated. As shown in Table V, compared with the baseline method, the reduction on

TABLE III
F-SCORE OF THE BASELINE IN RTO MODE

Class	Test ID						
	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Walking	93.8%	94.5%	91.9%	95.1%	82.5%	82.4%	95.4%
Jogging	90.3%	98.7%	97.3%	95.1%	89.1%	79.5%	94.6%
Sitting	89.1%	97.6%	99.7%	98.3%	96.3%	57.8%	96.2%
Standing	98.3%	98.0%	98.2%	97.8%	97.6%	85.9%	92.8%
Ascending stairs	33.1%	78.8%	74.6%	70.3%	65.9%	57.6%	63.2%
Descending stairs	55.0%	62.2%	70.0%	59.1%	51.7%	68.9%	68.4%
Average	76.6%	88.3%	88.6%	86.0%	80.5%	72.0%	85.1%

TABLE IV
F-SCORE OF TSE-CNN IN RTO MODE

Class	Test ID						
	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Walking	92.4% (-1.4%)	91.5% (-3.0%)	92.0% (+0.1%)	94.1% (-1.0%)	86.2% (+3.7%)	82.5% (+0.1%)	96.1% (+0.7%)
Jogging	92.5% (+2.2%)	99.5% (+0.8%)	98.5% (+1.2%)	96.8% (+1.7%)	96.1% (+7.0%)	83.4% (+3.9%)	97.3% (+2.7%)
Sitting	92.9% (+3.8%)	99.4% (+1.8%)	97.3% (-2.4%)	95.1% (-3.2%)	92.8% (-3.5%)	76.7% (+18.9%)	97.9% (+1.7%)
Standing	99.9% (+1.6%)	99.1% (+1.1%)	99.1% (+0.9%)	97.3% (-0.5%)	99.7% (+2.1%)	86.6% (+0.7%)	95.4% (+2.6%)
Ascending stairs	38.0% (+4.9%)	76.6% (-2.2%)	77.0% (+2.4%)	69.3% (-1.0%)	72.1% (+6.2%)	57.3% (-0.3%)	65.2% (+2.0%)
Descending stairs	59.2% (+4.2%)	60.3% (-1.9%)	72.6% (+2.6%)	69.6% (+10.5%)	54.1% (+2.4%)	80.2% (+11.3%)	74.2% (+5.8%)
Average	79.2% (+2.6%)	87.8% (-0.5%)	89.4% (+0.8%)	87.1% (+1.1%)	83.5% (+3.0%)	77.8% (+5.8%)	87.7% (+2.6%)

TABLE V
COMPARISON OF COMPUTATIONAL COMPLEXITY BETWEEN TSE-CNN AND THE BASELINE IN RTO MODE

	The Baseline	TSE-CNN		
		First-stage	Second-stage	Total
Parameters	14150	4355	4868	9223
No. Multiplication	731136	336384	336896	673280
No. addition	731136	336384	336896	673280

computational complexity for ascending stairs and descending stairs are 54% and 54% respectively. Assuming that ascending stairs and descending stairs account for about 10% of daily-life activities, the overall reduction on computational complexity is 13%, as calculate by (1), which will lead to proportional reduction on power consumption.

$$reduction = 1 - \frac{10\% \times N_f + 90\% \times (N_f + N_s)}{N_b} \quad (1)$$

In (1), N_f refers to the number of multiplication of the first-stage CNN and N_s refers to that of the second-stage CNN. N_b refers to that of the baseline. The reduction on computational complexity is calculated by considering that the second-stage CNN is inactive for 10% of the time (when ascending stairs and descending stairs).

B. Results for WTOD Mode

WTOD mode is used to evaluate the effect of the proposed data augmentation method. In this mode, a small amount (1/8) of the test subject's data is included in the training data. Table VI and Table VII shows the F-score of the baseline and the proposed TSE-CNN in the WTOD mode. It can be observed that the F-score of both is improved compared with RTO mode (Table III and Table IV) merely due to the inclusion of test subject's data (1/8) in the training data.

With the proposed data augmentation method, more artificial training data is generated. This helps further increase

the F-score. Table VIII shows the F-score of the proposed TSE-CNN with data augmentation method. Compared to TSE-CNN without data augmentation method (Table VII), the F-score is improved significantly, especially for ascending stairs and descending stairs (up to 6.3% and 6.0% respectively).

V. DISCUSSION

Table IX compares the proposed method with several state-of-the-art methods. For fair comparison, the proposed method is compared with methods targeting at similar activities including standing, sitting, walking, jogging, ascending stairs, descending stairs. As some of the methods use accuracy instead of F-score, we have also shown the accuracy of the proposed method for fair comparison. The table also details the information such as the type of sensors, the position of sensors, the datasets and the test mode of references. Most of benchmarked work used 1-2 sensors (e.g., accelerometer, gyroscope) placed at the test subjects' body such as pants pocket and waist.

In the RTO mode, in terms of accuracy, the proposed TSE-CNN is the highest, followed by the baseline. In terms of the F-score, the proposed TSE-CNN is the highest for most activities except for walking (second only to the baseline). Especially for ascending stairs and descending stairs, the proposed TSE-CNN has significantly higher F-score than other methods. In the WTOD mode, the proposed TSE-CNN is higher than baseline for both accuracy and F-score. When combining the proposed TSE-CNN and data augmentation method, a larger improvement is achieved, especially for ascending stairs and descending stairs. More importantly, these are achieved by using data from a single accelerometer, while some of the benchmarked methods use data from multiple sensors. The proposed method is able to achieve an average recognition accuracy of 87.9% for RTO mode and 95.7% for WTOD mode. Moreover, the computation complexity for ascending/descending stairs can be reduced by 54% as shown in Section IV.A.

TABLE VI
F-SCORE OF THE BASELINE IN WTOD MODE

Class	Test ID						
	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Walking	98.1%	96.9%	97.3%	98.3%	96.3%	97.7%	98.5%
Jogging	94.9%	99.4%	98.2%	96.8%	95.1%	94.9%	98.5%
Sitting	99.6%	99.5%	99.9%	98.5%	98.8%	86.8%	98.9%
Standing	99.9%	100%	99.4%	98.3%	99.3%	90.6%	98.0%
Ascending stairs	67.1%	84.8%	86.5%	87.7%	80.9%	78.1%	81.9%
Descending stairs	85.7%	75.5%	82.0%	81.1%	72.2%	85.2%	82.9%
Average	90.9%	92.7%	93.9%	93.4%	90.4%	88.9%	93.1%

TABLE VII
F-SCORE OF TSE-CNN IN WTOD MODE

Class	Test ID						
	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Walking	98.4%	97.1%	97.5%	98.3%	96.9%	97.0%	98.5%
Jogging	95.9%	99.4%	98.7%	98.4%	97.7%	95.5%	99.1%
Sitting	99.0%	99.4%	96.8%	98.6%	99.5%	99.4%	99.9%
Standing	99.9%	100%	98.9%	98.1%	99.9%	98.7%	99.0%
Ascending stairs	73.9%	85.0%	89.8%	88.0%	81.5%	77.3%	82.4%
Descending stairs	89.8%	75.9%	83.6%	85.6%	72.4%	86.1%	85.4%
Average	92.8%	92.8%	94.2%	94.5%	91.3%	92.4%	94.0%

TABLE VIII
F-SCORE OF TSE-CNN WITH DATA AUGMENTATION IN WTOD MODE

Class	Test ID						
	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Walking	98.7% (+0.3%)	98.1% (+1.0%)	98.0% (+0.5%)	98.6% (+0.3%)	97.9% (+1.0%)	97.1% (+0.1%)	98.8% (+0.3%)
Jogging	96.6% (+0.7%)	99.7% (+0.3%)	98.7% (+0.0%)	98.6% (+0.2%)	96.9% (-0.8%)	95.8% (+0.3%)	99.5% (+0.4%)
Sitting	98.1% (+0.9%)	99.5% (+0.1%)	99.3% (+2.5%)	98.6% (+0.0%)	98.6% (-0.9%)	99.7% (+0.3%)	99.9% (+0.0%)
Standing	99.9% (+0.0%)	99.9% (-0.1%)	99.1% (+0.2%)	98.4% (+0.3%)	99.2% (-0.7%)	98.7% (+0.0%)	99.2% (+0.2%)
Ascending stairs	74.5% (+0.6%)	86.0% (+1.0%)	89.8% (+0.0%)	87.9% (-0.1%)	84.1% (+2.6%)	77.1% (-0.2%)	88.7% (+6.3%)
Descending stairs	89.7% (-0.1%)	81.9% (+6.0%)	86.7% (+3.1%)	87.3% (+1.7%)	78.3% (+5.9%)	87.8% (+1.7%)	87.3% (+1.9%)
Average	92.9% (+0.1%)	94.2% (+1.4%)	95.3% (+1.1%)	94.9% (+0.4%)	92.5% (+1.2%)	92.7% (+0.3%)	95.6% (+1.6%)

TABLE IX
COMPARISON OF ACCURACY & F-SCORE WITH THE STATE OF THE ART METHODS

	[8] IEEE Access 2017	[10] EUSIPCO 2017	[15] IDEAL 2015	[26] EMBC 2016	[36] IEEE J-BHI 2018	[37] IEEE J-BHI 2018	The Baseline		TSE-CNN		TSE-CNN +Aug
							RTO	WTOD	RTO	WTOD	WTOD
Accuracy	-	85.1%	85.4% ⁺	81%	84.6%		86.4%	94.3%	87.9%	95.1%	95.7%
Average F-score	67.5% ⁺	-	81.2% ⁺	-	-	80.9%	82.5%	91.9%	84.6%	93.2%	94.0%
Walking F-score	-	-	89.6% ⁺	85.9% ⁺	82.9% ⁺	85.7%	90.8%	97.6%	90.6%	97.7%	98.2%
Jogging F-score	-	-	-	-	-	-	92.1%	96.8%	94.9%	97.8%	98.0%
Sitting F-score	-	-	91.4% ⁺	85.0% ⁺	94.2% ⁺	82.6%	90.7%	97.4%	93.2%	98.9%	99.1%
Standing F-score	-	-	90.5% ⁺	78.3% ⁺	83.2% ⁺	86.3%	95.5%	97.9%	96.7%	99.2%	99.2%
Ascending stairs F-score	-	-	55.3% ⁺	4.9% ⁺	-	44.2%	63.4%	81.0%	65.1%	82.6%	84.0%
Descending stairs F-score	-	-	66.8% ⁺	25.0% ⁺	-	64.0%	62.2%	82.7%	67.2%	82.7%	85.6%
Sensors [*]	1A & 1G	1A & 1G & 1M	1A	1A	2A	1A	1A				
Position [#]	P	HA / P / BA	P	WA	N & T	WR	P				
Dataset	SC ^{**}	SC ^{**}	WISDM	SC ^{**}	SC ^{**}	PAMAP2	WISDM				
Test mode	RTO	RTO	RTO	RTO	WTOD	RTO	RTO & WTOD				

⁺ The values are calculate using the confusion matrix, precision or recall provided in the papers.

^{*} A refers to accelerometer. G refers to gyroscope. M refers to magnetometer. The number refers to the number of each type of sensor.

[#] P refers to pants pocket. WA refers to waist. N refers to neck. WR refers to wrist. T refers to thigh. HA refers to hand. BA refers to bag. The symbol "&" refers to AND. The symbol "/" refers to OR.

^{**} Self Collected Dataset.

VI. CONCLUSION

The existing HAR methods (including neural network based methods and other methods) suffer from high computational complexity and low recognition accuracy especially for activities with less training data and more complex behavior pattern. This paper proposes a new HAR method with a two-stage end-to-end CNN structure and a data augmentation method. Compared with the state-of-the-art HAR methods, the proposed method achieves significantly improved recognition accuracy and reduced computational complexity. It is able to achieve an average recognition accuracy of 87.9% for RTO mode and 95.7% for WTOD mode. The computation complexity for ascending/descending stairs can be reduced by 54%. The proposed methods are suitable for sensor-based wearable HAR requiring high accuracy and low power consumption.

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